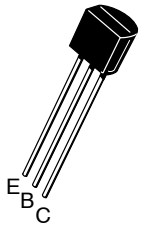
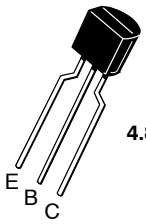
NPN General-Purpose Amplifier

2N5551



TO-92-3
4.825x4.76
CASE 135AN



TO-92-3
4.83x4.76 LEADFORMED
CASE 135AR

Description

This device is designed for general-purpose high-voltage amplifiers and gas discharge display drivers.

Features

- These Devices are Pb-Free, Halogen Free/BFR Free and are RoHS Compliant

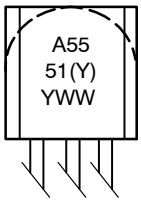
ABSOLUTE MAXIMUM RATINGS (Note 1)

Symbol	Parameter	Value	Unit
V_{CEO}	Collector-Emitter Voltage	160	V
V_{CBO}	Collector-Base Voltage	180	V
V_{EBO}	Emitter-Base Voltage	6	V
I_C	Collector Current – Continuous	600	mA
T_J, T_{STG}	Operating and Storage Temperature (Note 2)	–55 to + 150	°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. These ratings are limiting values above which the serviceability of any semiconductor device may be impaired.
2. These ratings are based on a maximum junction temperature of 150 °C. These are steady-state limits. **onsemi** should be consulted on applications involving pulsed or low-duty cycle operations.

MARKING DIAGRAM



A = Assembly Location
5551(Y) = Specific Device Code
Y = Year
WW = Work Week

ORDERING INFORMATION

See detailed ordering and shipping information on page 5 of this data sheet.

NOTE: Some of the devices on this data sheet have been **DISCONTINUED**. Please refer to the table on page 5.

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THERMAL CHARACTERISTICS (T_A = 25°C unless otherwise noted) (Note 3)

Symbol	Characteristic	Max	Unit
P_D	Total Device Dissipation	625	mW
	Derate Above 25 °C	5.0	mW/°C
$R_{\theta JC}$	Thermal Resistance, Junction to Case	83.3	°C/W
$R_{\theta JA}$	Thermal Resistance, Junction to Ambient	200	°C/W

ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted) (Note 4)

Symbol	Parameter	Test Conditions	Min	Max	Unit
$V_{(BR)CEO}$	Collector-Emitter Breakdown Voltage	I_C = 1.0 mA, I_B = 0	160		V
$V_{(BR)CBO}$	Collector-Base Breakdown Voltage	I_C = 100 μA, I_E = 0	180		V
$V_{(BR)EBO}$	Emitter-Base Breakdown Voltage	I_E = 10 μA, I_C = 0	6.0		V
I_{CBO}	Collector Cut-Off Current	V_{CB} = 120 V, I_E = 0		50	nA
		V_{CB} = 120 V, I_E = 0 V, T_A = 100 °C		50	μA
I_{EBO}	Emitter Cut-Off Current	V_{EB} = 4.0 V, I_C = 0		50	nA
ON CHARACTERISTICS					
h_{FE}	DC Current Gain	I_C = 1.0 mA, V_{CE} = 5.0 V	80		
		I_C = 10 mA, V_{CE} = 5.0 V	80	250	
		I_C = 10 mA, V_{CE} = 5.0 V (for 2N5551YBU, 2N5551YTA)	180	240	
		I_C = 50 mA, V_{CE} = 5.0 V	30		
$V_{CE(sat)}$	Collector-Emitter Saturation Voltage	I_C = 10 mA, I_B = 1.0 mA		0.15	V
		I_C = 50 mA, I_B = 5.0 mA		0.20	V
$V_{BE(sat)}$	Base-Emitter On Voltage	I_C = 10 mA, I_B = 1.0 mA		1.0	V
		I_C = 50 mA, I_B = 5.0 mA		1.0	V
SMALL-SIGNAL CHARACTERISTICS					
f_T	Current Gain Bandwidth Product	I_C = 10 mA, V_{CE} = 10 V, f = 100 MHz	100		MHz
C_{obo}	Output Capacitance	V_{CB} = 10 V, I_E = 0, f = 1.0 MHz		6.0	pF
C_{ibo}	Input Capacitance	V_{BE} = 0.5 V, I_C = 0, f = 1.0 MHz		20	pF
H_{fe}	Small-Signal Current Gain	I_C = 1.0 mA, V_{CE} = 10 V, f = 1.0 kHz	50	250	
NF	Noise Figure	I_C = 250 μA, V_{CE} = 5.0 V, R_S = 1.0 kΩ, f = 10 Hz to 15.7 kHz		8.0	dB

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

3. PCB board size FR-4 76 x 114 x 0.6 T mm^3 (3.0 inch x 4.5 inch x 0.062 inch) with minimum land pattern size.
4. Pulse test: pulse width ≤ 300 μs, duty cycle ≤ 2.0%.

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TYPICAL PERFORMANCE CHARACTERISTICS

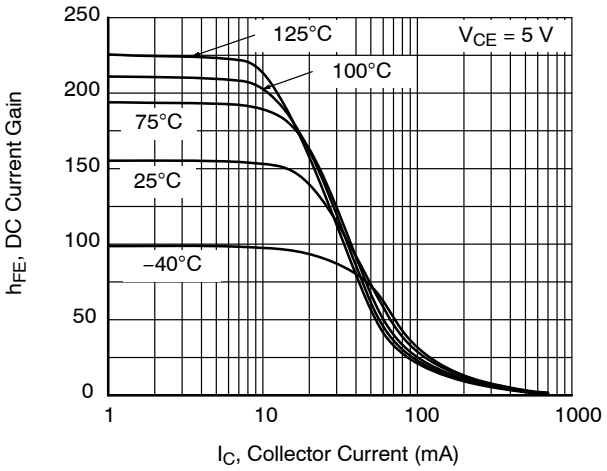


Figure 1. Typical Pulsed Current Gain vs. Collector Current

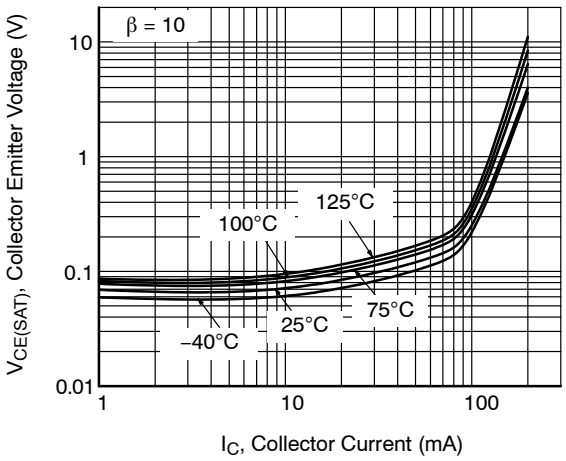


Figure 2. Collector- Emitter Saturation Voltage vs. Collector Current

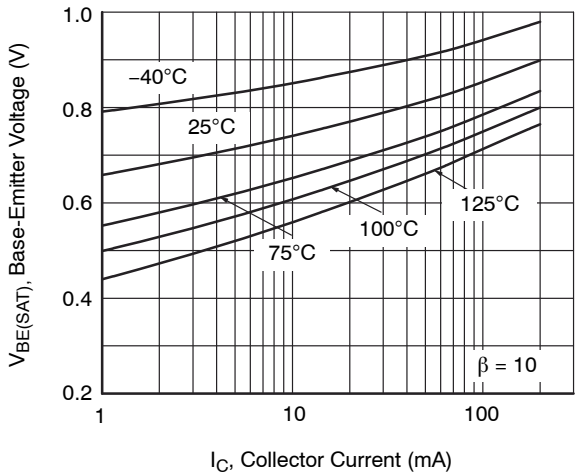


Figure 3. Base-Emitter Saturation Voltage vs. Collector Current

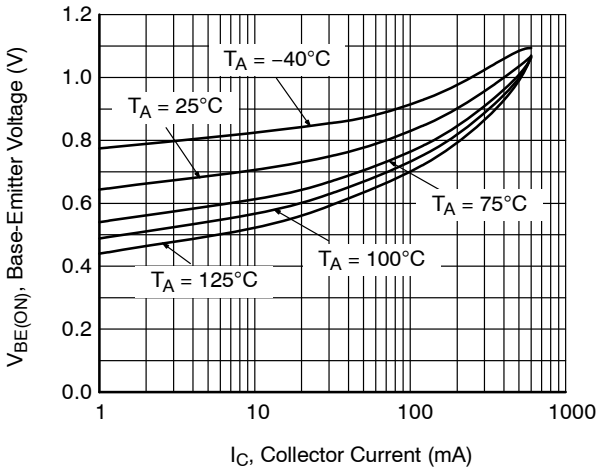


Figure 4. Base-Emitter On Voltage vs. Collector Current

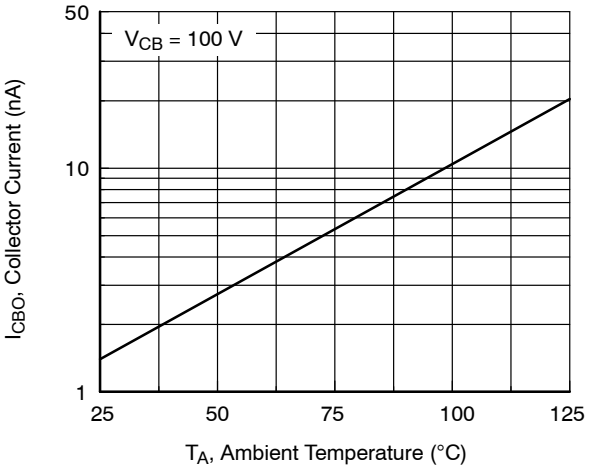


Figure 5. Collector Cut-Off Current vs. Ambient Temperature

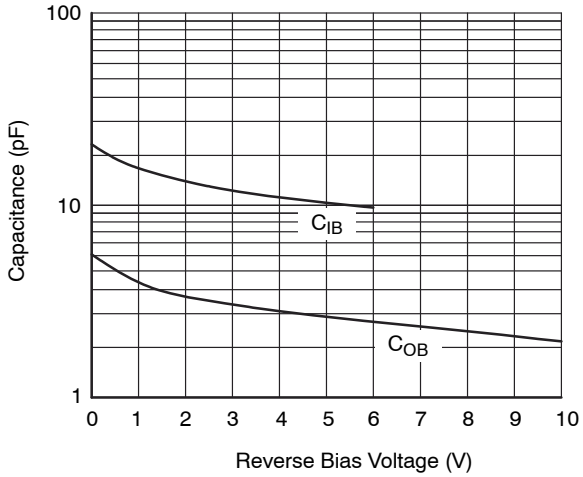


Figure 6. Input and Output Capacitance vs. Reverse Voltage

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TYPICAL PERFORMANCE CHARACTERISTICS (continued)

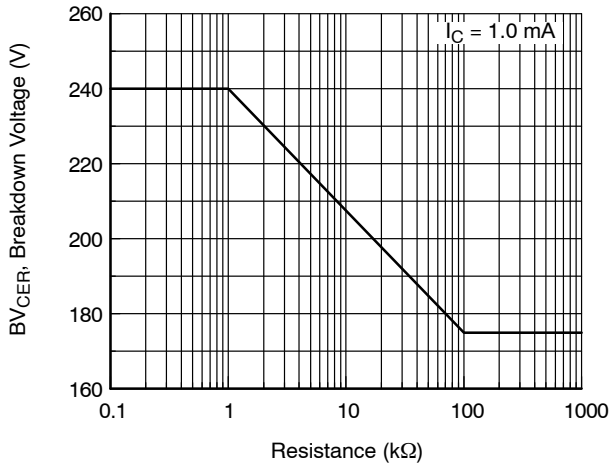


Figure 7. Collector-Emitter Breakdown Voltage with Resistance between Emitter-Base

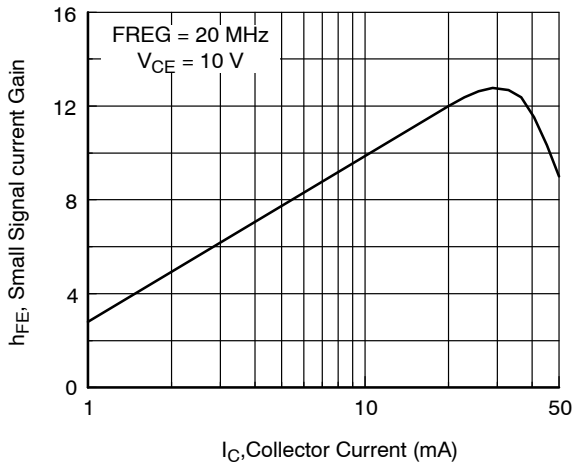


Figure 8. Small Signal Current Gain vs. Collector Current

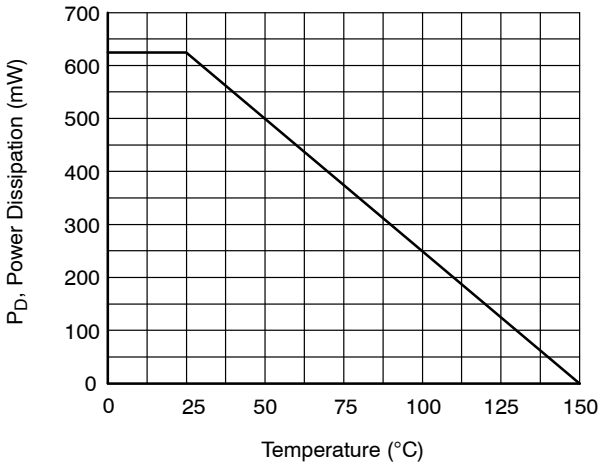


Figure 9. Power Dissipation vs. Ambient Temperature

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ORDERING INFORMATION (Note 5)

Part Number	Top Mark	Package	Shipping†
2N5551TA	5551	TO-92-3, Case 135AR (Pb-Free)	2000 / Ammo Pack
2N5551TFR	5551	TO-92-3, Case 135AR (Pb-Free)	2000 / Tape & Reel
2N5551TF	5551	TO-92-3, Case 135AR (Pb-Free)	2000 / Tape & Reel
2N5551BU	5551	TO-92-3, Case 135AN (Pb-Free)	10000 / Bulk Bag

DISCONTINUED (Note 6)

2N5551CTA	5551	TO-92-3, Case 135AR (Pb-Free)	2000 / Ammo Pack
2N5551YTA	5551Y	TO-92-3, Case 135AR (Pb-Free)	2000 / Ammo Pack
2N5551YBU	5551Y	TO-92-3, Case 135AN (Pb-Free)	10000 / Bulk Bag

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

5. Suffix "-Y" means h_{FE} 180~240 in 2N5551 (Test condition: I_C = 10 mA, V_{CE} = 5.0 V)
6. **DISCONTINUED:** These devices are not available. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.

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REVISION HISTORY

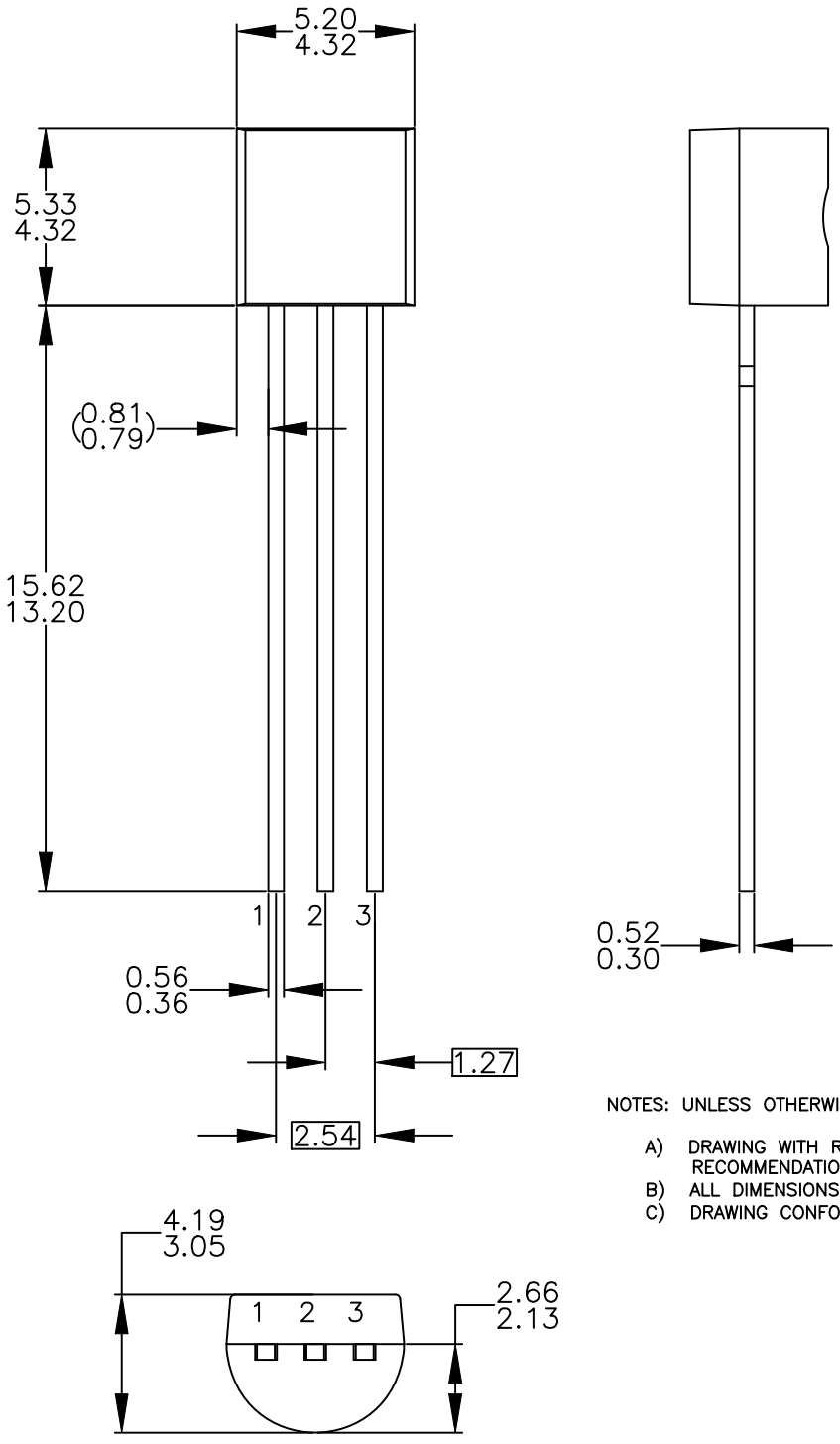
Revision	Description of Changes	Date
7	2N5551YBU OPN Marked as Discontinued.	1/14/2026

This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

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TO−92 3 4.825x4.76
CASE 135AN
ISSUE O

DATE 31 JUL 2016



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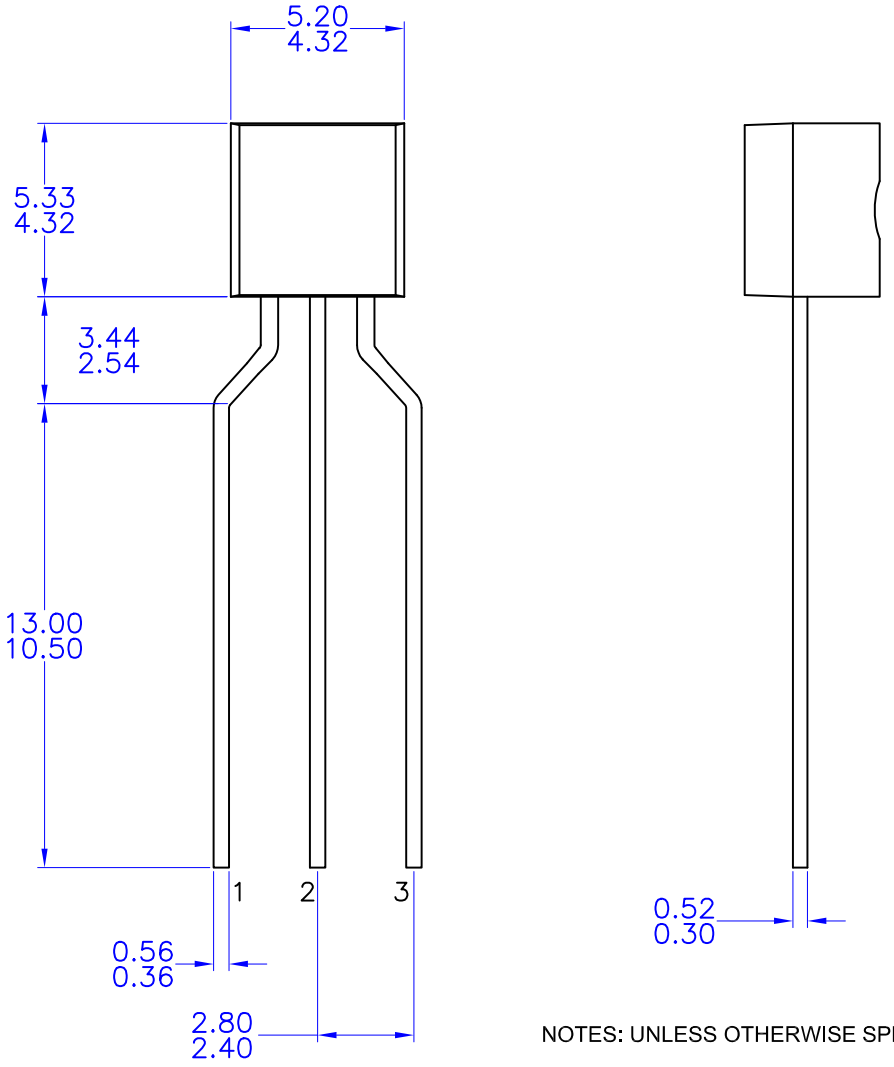
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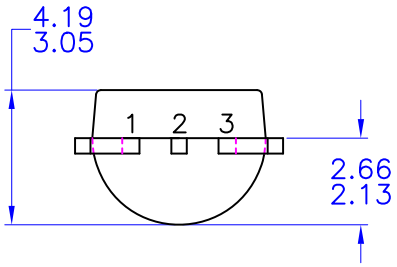
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CASE 135AR
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